Choose an equation for steady acceleration

LEARNING TARGET · R5 I can choose an equation for constant acceleration.

A · KEY VOCABULARY

Steady a	Equal velocity changes in equal times.
Knowns	List supplied values before choosing.
Displacement	Signed change in position, not distance.
Cross-check	Test with units, a graph or another equation.

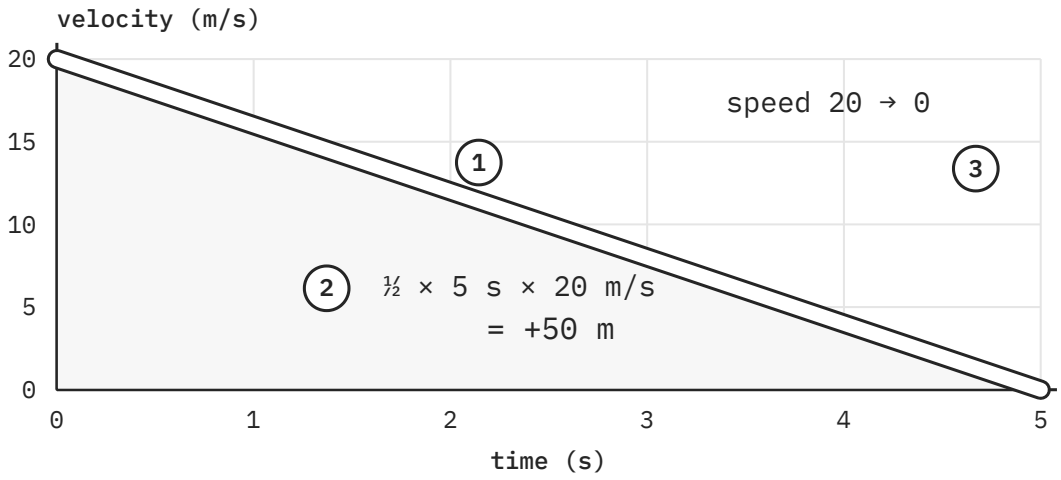
B · KEY EQUATIONS + UNITS

Equation	Omits
$v_f = v_i + at$	displacement
$\Delta x = v_i t + \frac{1}{2}at^2$	final velocity
$v_f^2 = v_i^2 + 2a\Delta x$	time
$\Delta x = \frac{1}{2}(v_i + v_f)t$	acceleration

Use for motion along one axis with constant acceleration.

C · COLOR IT · CHECK WITH AREA

Straight line: steady a.



☐ **1 · PURPLE** Velocity line

☐ **2 · BLUE** Displacement area

☐ **3 · RED** Decreasing speed

D · WORKED EXAMPLE

A car moving at +**20 m/s** brakes at **−4 m/s²** until it stops. **Find its stopping distance, then check the result.**

GIVEN	EQUATION	WORK	ANSWER
v_i = +**20 m/s** v_f = **0 m/s** a = **−4 m/s²**	No time given: $v_f^2 = v_i^2 + 2a\Delta x$.	$0 = 400 - 8\Delta x$ $t = (0 - 20)/(-4)$ area = ½(20)(5)	Δx = **+50 m** **5 s** to stop; area also gives **50 m**.

E · THE 2026-XJ THREAD

DAY 8 OF 22 · NEW TOOL: EQUATION CHOICE

For XJ, choose a relationship that fits both the known values and the motion being assumed. A graph provides a second check before a forecast is reported.

Unit question: How fast is it moving, and where will it arrive?

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else